

What is ModBus TCP ?



Modbus TCP is a network communication protocol used in industrial automation to allow devices like PLCs, HMIs, sensors, drives, and SCADA systems to exchange data over an Ethernet network.

It combines:

- **Modbus:** An application-layer protocol originally developed by Modicon in 1979.
- **TCP/IP:** The standard networking protocol used on Ethernet.

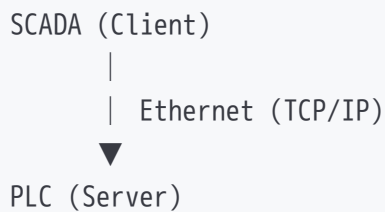
How it works

Modbus TCP uses a **client-server** model (formerly called master-slave):

- **Client:** Initiates requests (e.g., a PLC or SCADA software).
- **Server:** Responds to requests (e.g., an I/O module, energy meter, or sensor).

For example:

text



The client asks for information, and the server replies.

Default Port

Modbus TCP uses **TCP port 502**.

What data can it read or write?

Modbus organizes data into four main types:

Data Type	Access	Typical Use
Coils	Read/Write	Digital outputs
Discrete Inputs	Read Only	Digital inputs

Data Type	Access	Typical Use
Input Registers	Read Only	Sensor measurements
Holding Registers	Read/Write	Configuration values, analog values

For example:

- Holding Register 40001 = Motor speed
- Coil 00010 = Start/Stop command
- Input Register 30001 = Temperature

Example communication

Suppose a temperature sensor stores its value in Holding Register 40001.

1. Client sends:

Read Holding Registers starting at address 0, quantity 1

2. Server replies:

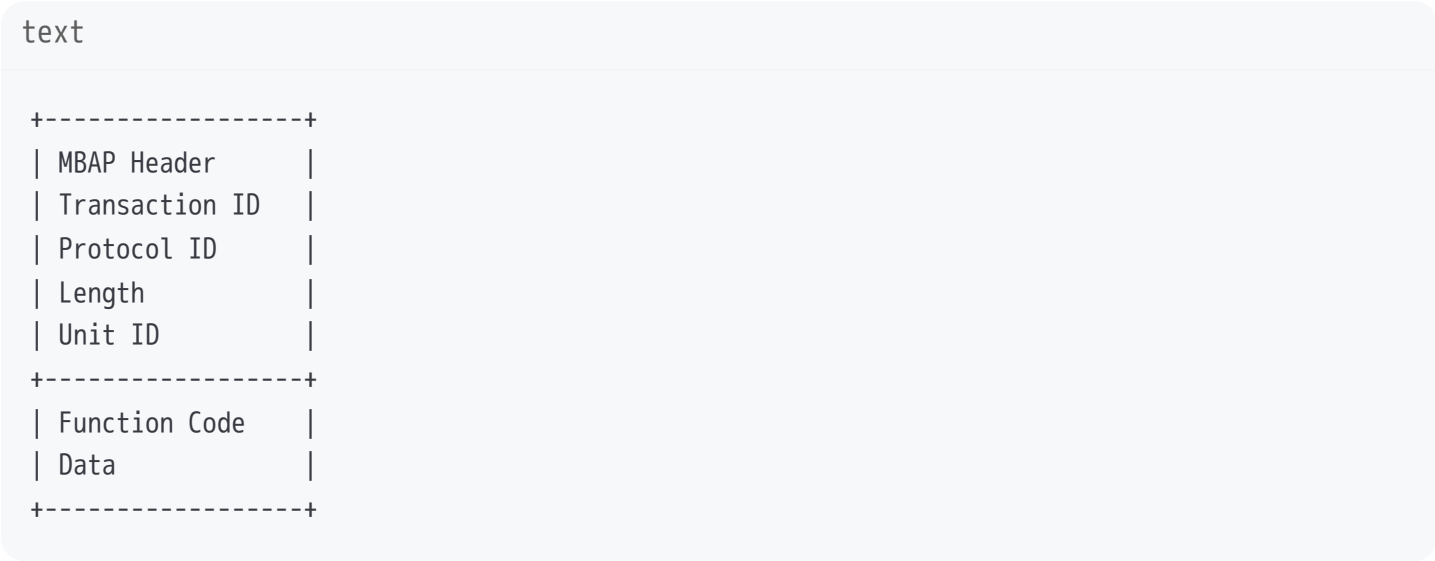
Register value = 250

If the value is scaled by 10:

250 = 25.0°C

Modbus TCP packet structure

A Modbus TCP message contains:



Unlike Modbus RTU, there is **no CRC checksum** because TCP/IP already provides error checking.

Common function codes

Function Code	Description
01	Read Coils
02	Read Discrete Inputs
03	Read Holding Registers
04	Read Input Registers
05	Write Single Coil
06	Write Single Register
15	Write Multiple Coils
16	Write Multiple Registers

Modbus TCP vs. Modbus RTU

Feature	Modbus TCP	Modbus RTU
Network	Ethernet	RS-232/RS-485
Transport	TCP/IP	Serial
Default Port	502	N/A
Speed	10 Mbps to multi-Gbps Ethernet	Typically 9.6 kbps–115.2 kbps
Error Checking	TCP checksum	CRC-16
Topology	Switched Ethernet	Multi-drop serial bus
Multiple Clients	Supported	Typically one master

Advantages

- Uses standard Ethernet infrastructure.
- Higher communication speeds than serial Modbus.
- Easier integration with IT networks and SCADA systems.
- Supports multiple simultaneous client connections (depending on the server implementation).
- Widely supported by industrial equipment from many manufacturers.

Typical applications

Modbus TCP is commonly used for:

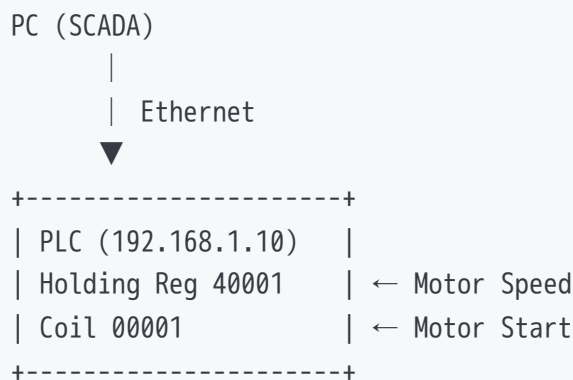
- PLC-to-HMI communication

- SCADA data collection
- Energy meters
- Variable Frequency Drives (VFDs)
- Remote I/O modules
- Temperature and pressure transmitters
- Building automation systems (HVAC)
- Industrial robots and machine monitoring

Simple example

Imagine a PLC connected to a motor:

text



The SCADA application can:

- Read Holding Register 40001 to display the motor speed.
- Write Coil 00001 to start or stop the motor.

In short, **Modbus TCP is one of the most widely used industrial Ethernet protocols because it is simple, open, and interoperable, making it a common choice for communication between automation devices from different manufacturers.**